

IT3010 - NDM

Practical 01- Lab Report



IT Number	Name	Email	Batch	Provider
IT23199576	K.H.M.D.M.Herath	IT23199576@my.sliit.lk	WE_IT_Y3_S2	VMware Workstation

Introduction

Virtualization is a technology that allows multiple operating systems to run on a single physical computer by abstracting hardware resources. Instead of dedicating one physical machine to one operating system, virtualization enables the creation of Virtual Machines (VMs), each functioning as an independent system.

At the core of virtualization is the Virtual Machine Monitor (VMM), also known as the hypervisor. The hypervisor manages hardware resources such as CPU, memory, storage, and networking, and ensures isolation between virtual machines.

One of the most important components in virtualization is virtual networking. Virtual Network Interface Cards (vNICs) allow virtual machines to communicate with the host machine, other virtual machines, and external networks. VMware Workstation provides different types of virtual network adapters to support various networking and security requirements.

This report focuses on VMware Workstation and analyzes the types of virtual network adapters it provides, compares them, and explains practical usage scenarios.

Provide 01 – VMware Workstation

1. Provider -

VMware Workstation is a Type 2 hypervisor designed for desktop virtualization. It allows users to run multiple operating systems such as Windows and Linux on a single physical computer.

It is widely used for:

- Software development and testing
- Network simulation and lab environments
- Security testing
- Learning and research

VMware Workstation provides flexible networking options through different virtual network adapters.

2. Types of Virtual Network Interface Cards (NICs) in VMware

a) Bridged Network

In Bridged mode, the virtual machine is directly connected to the physical network. The VM behaves like a separate physical computer on the same LAN.

Characteristics:

- The VM receives its own IP address from the network's DHCP server.
- It is visible to other devices on the network.
- It can communicate with routers, switches, and other computers.

Advantages:

- Suitable for server testing.
- Allows full LAN access.
- Realistic network simulation.

Disadvantages:

- Lower security compared to other modes.
- VM is exposed to the network.

b) NAT (Network Address Translation)

In NAT mode, the virtual machine shares the host computer's IP address to access external networks.

Characteristics:

- The VM can access the internet.
- The VM is hidden from other devices on the LAN.
- IP address is assigned internally by VMware.

Advantages:

- Provides internet access.
- Prevents external access to the VM.
- Avoids IP conflicts.

Disadvantages:

- VM cannot be directly accessed from outside without additional configuration.

c) Host-Only Network

In Host Only mode, the virtual machine can communicate only with the host machine and other VMs connected to the same host only network.

Characteristics:

- No internet access.
- Fully isolated from external networks.
- Suitable for secure testing environments.

Advantages:

- High security.
- Safe environment for malware testing or network experiments.

Disadvantages:

- No external connectivity.

3. Comparison of VMware Network Adapters

Feature	Bridged	NAT	Host-Only
Internet Access	Yes	Yes	No
Visible to LAN	Yes	No	No
VM ↔ Host Communication	Yes	Yes	Yes
VM ↔ VM Communication	Yes	Yes	Yes
Security Level	Low	Medium	High
IP Source	Network DHCP	VMware NAT	VMware Internal

4. Practical Scenario (VMware – NAT)

Scenario:

A university student installs Ubuntu Linux in VMware Workstation to practice programming. The student needs internet access to download software packages and updates.

The student selects NAT mode so the VM can access the internet while remaining hidden from other devices on the campus network. This ensures both connectivity and security.

5. Reason for Using Different Adapters in VMware

VMware provides different virtual network adapters to:

- Support different security levels
- Allow flexible network configurations
- Simulate real-world network environments
- Enable safe and isolated testing
- Avoid IP conflicts

Different networking modes ensure that users can select the most appropriate configuration based on their requirements.

6. Conclusion

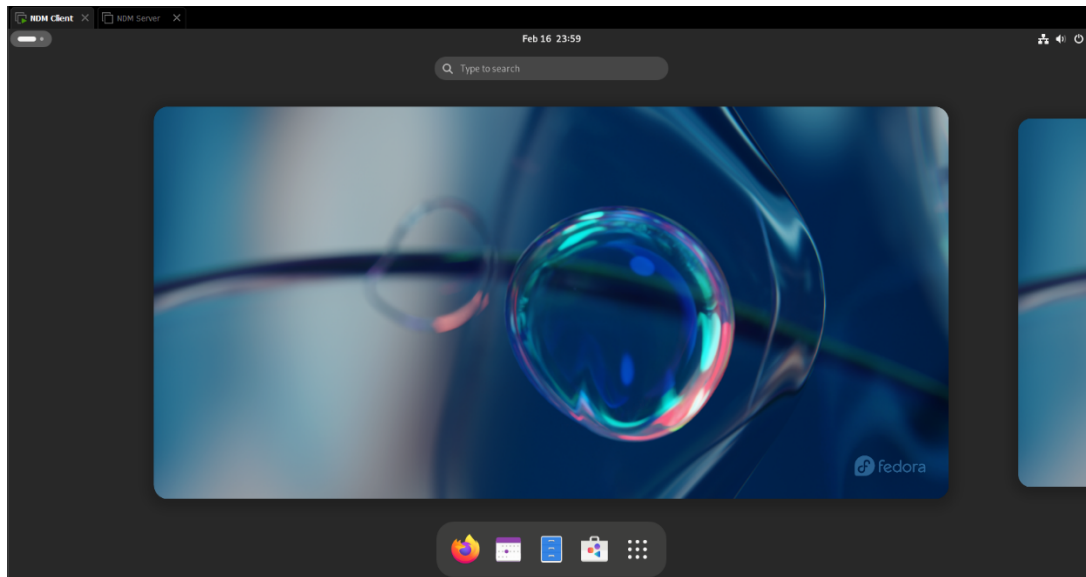
VMware Workstation provides powerful and flexible virtual networking capabilities through its Bridged, NAT, and Host-Only adapters. Each networking mode serves a specific purpose depending on the user's requirements for internet access, security, and LAN visibility.

Bridged mode is suitable for full network participation, NAT provides safe internet access, and Host-Only ensures complete isolation. Understanding these networking options is essential for effective network design, testing, and management in virtual environments.

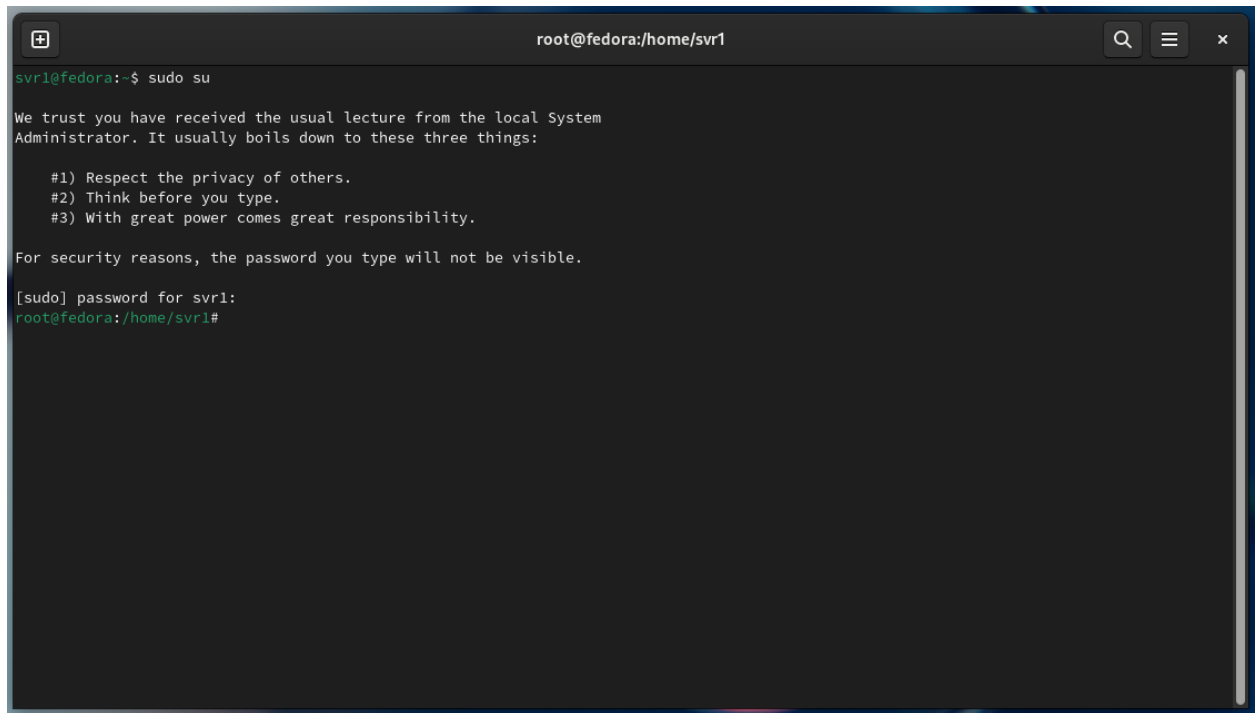
Through this practical study, it is clear that selecting the appropriate virtual network adapter is crucial for achieving desired connectivity and security outcomes in VMware Workstation.

Practical 1: Client Configuration

1. Go to fedora OS



2. Open the terminal and type “sudo su” and type password for go root



A terminal window titled "root@fedora:/home/svr1" with search, menu, and close icons in the top right. The prompt is "svr1@fedora:~\$". The user enters "sudo su". The terminal displays a message from the local System Administrator about security and responsibility, followed by a password prompt "[sudo] password for svr1:". After the password is entered, the prompt changes to "root@fedora:/home/svr1#".

```
svr1@fedora:~$ sudo su

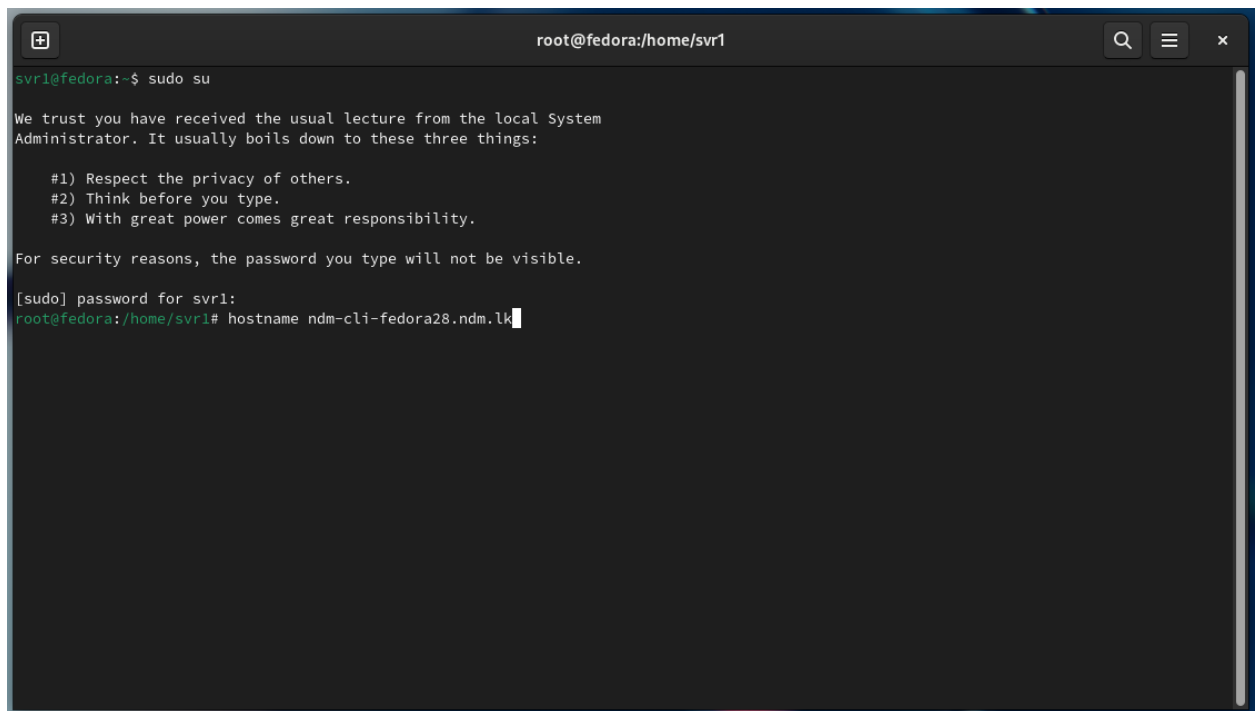
We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for svr1:
root@fedora:/home/svr1#
```

3. Change hostname



A terminal window titled "root@fedora:/home/svr1" with search, menu, and close icons in the top right. The prompt is "svr1@fedora:~\$". The user enters "sudo su". The terminal displays the same message from the local System Administrator as in the previous screenshot. After the password is entered, the prompt is "root@fedora:/home/svr1#". The user then enters the command "hostname ndm-cli-fedora28.ndm.lk".

```
svr1@fedora:~$ sudo su

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

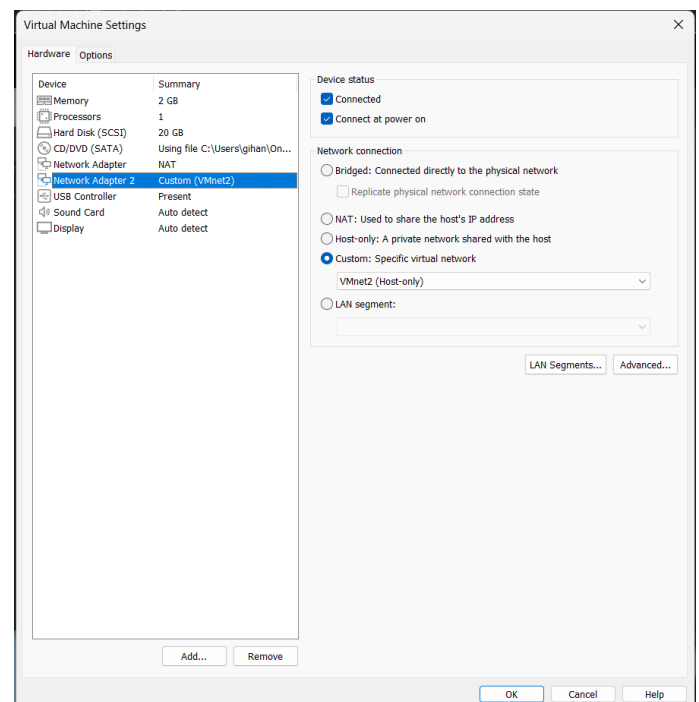
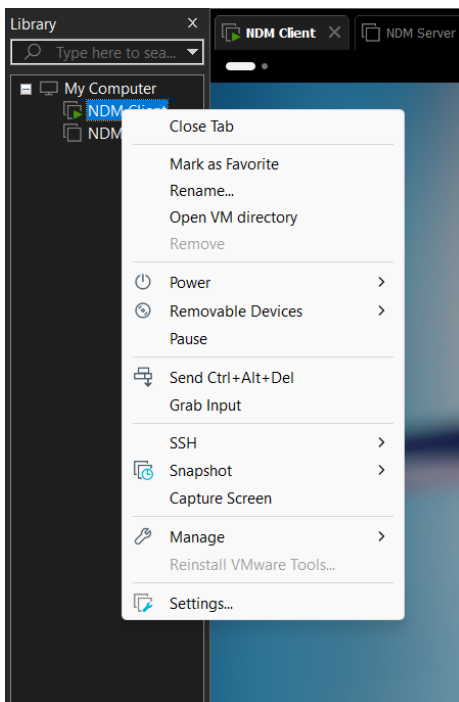
For security reasons, the password you type will not be visible.

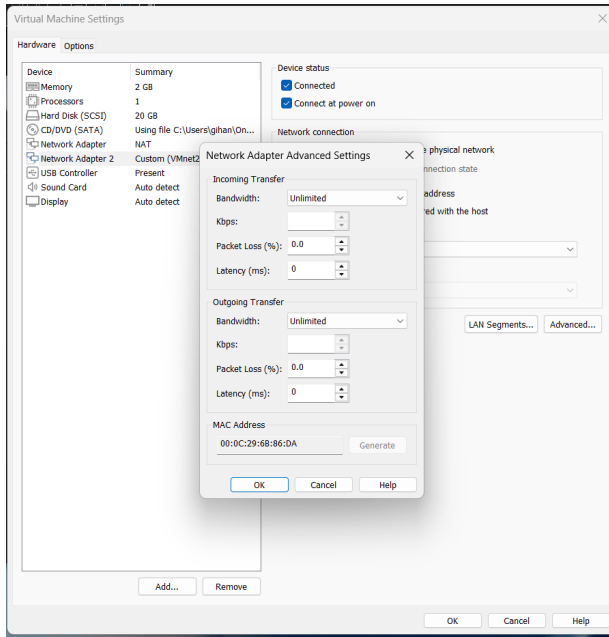
[sudo] password for svr1:
root@fedora:/home/svr1# hostname ndm-cli-fedora28.ndm.lk
```

4. Check IP addresses

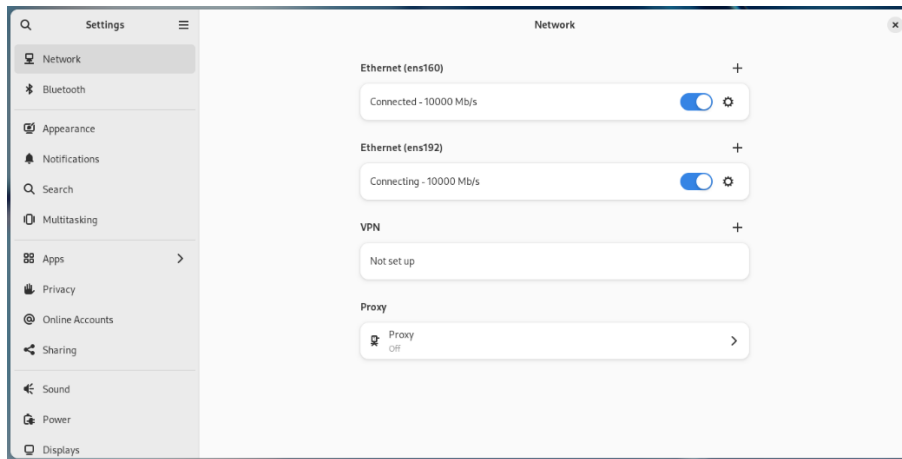
```
root@fedora:/home/svr1
svr1@fedora:~$ sudo su
[sudo] password for svr1:
root@fedora:/home/svr1# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:6b:86:d0 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.234.129/24 brd 192.168.234.255 scope global dynamic noprefixroute ens160
        valid_lft 1298sec preferred_lft 1298sec
    inet6 fe80::51a7:bfed:4995:4f31/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens192: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:6b:86:da brd ff:ff:ff:ff:ff:ff
    altname enp11s0
    inet 10.0.1.3/24 brd 10.0.1.255 scope global noprefixroute ens192
        valid_lft forever preferred_lft forever
    inet6 fe80::111f:2e3:9adc:cfe2/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
root@fedora:/home/svr1#
```

5. Go to the settings and check MAC address





- After checking mac address, go to the FedoraOS settings and check what is the same Mac address for the network adapter

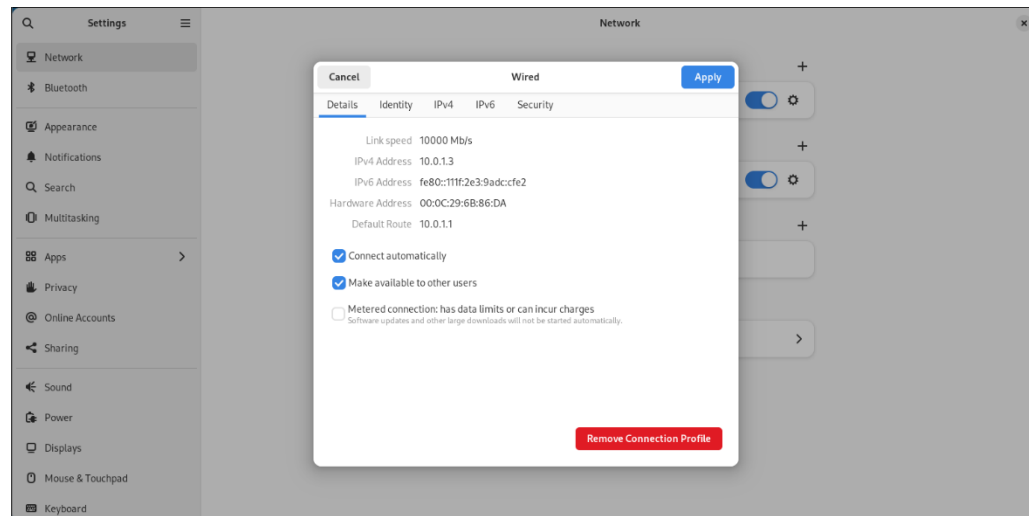
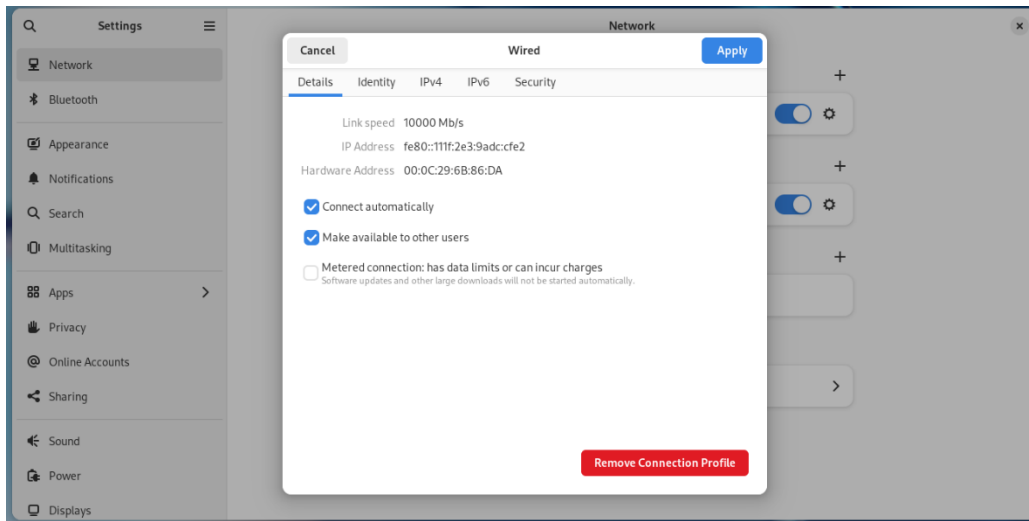


Ethernet (ens192)

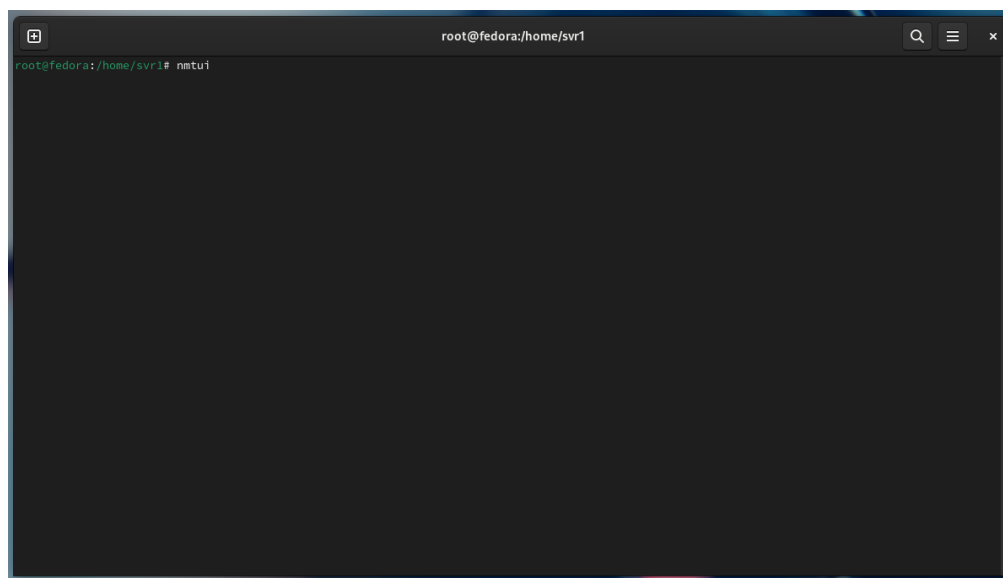


Connecting - 10000 Mb/s

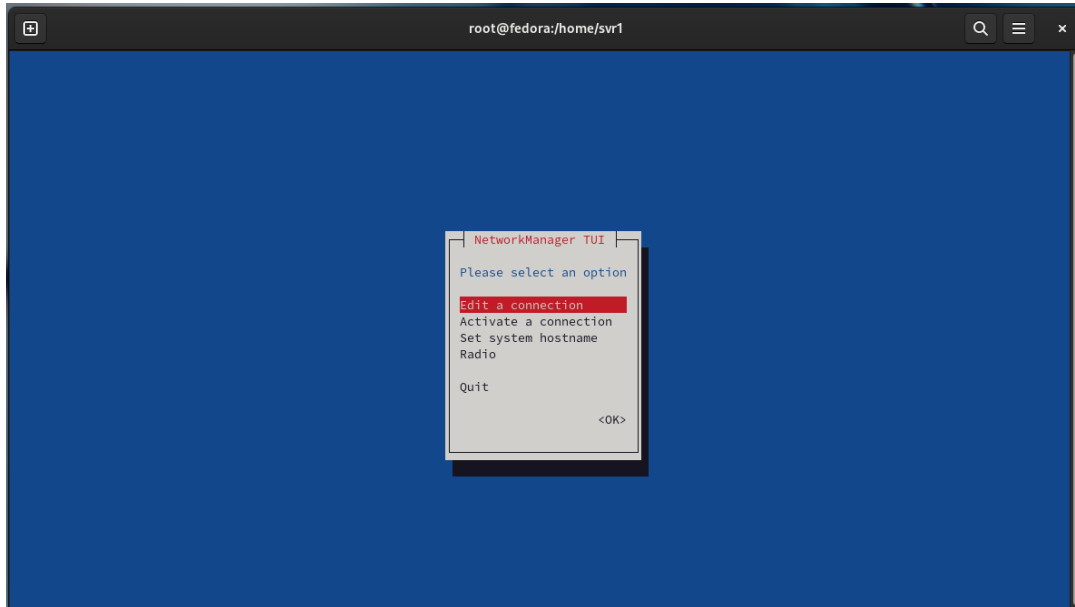




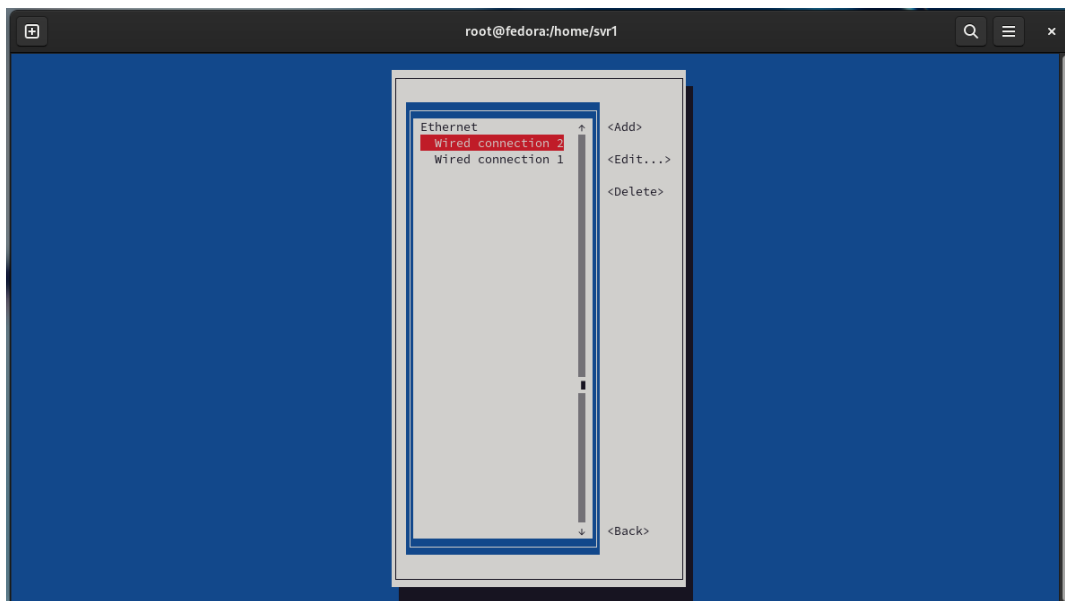
7. Confirm the mac address of network adapter, then start configurations.
8. Open terminal and go to the root (2nd step) type “nmtui” for network configurations

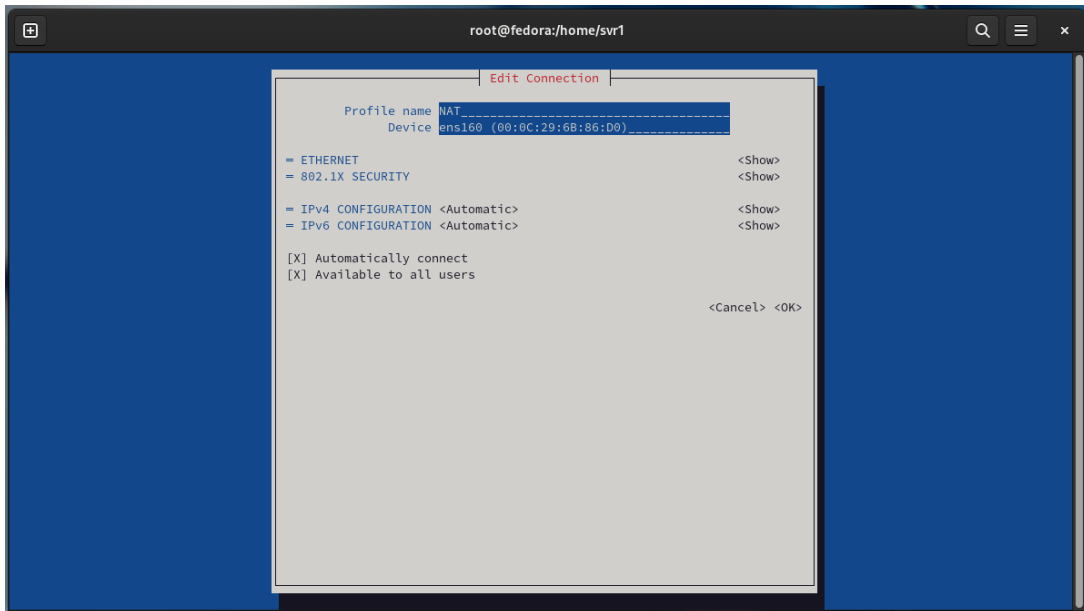
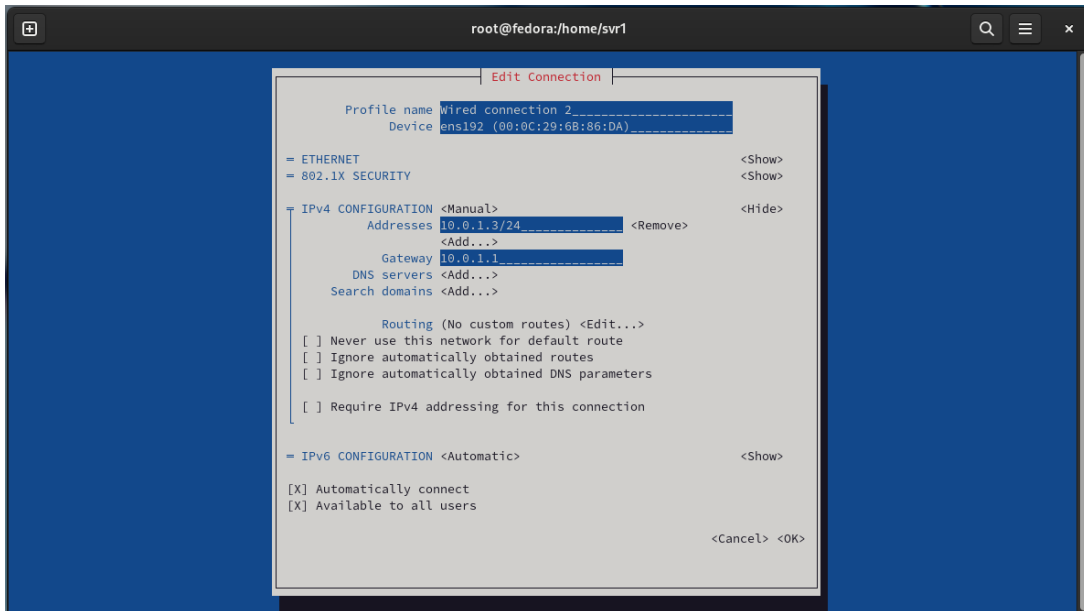


9. After that go to the “Edit a connection”

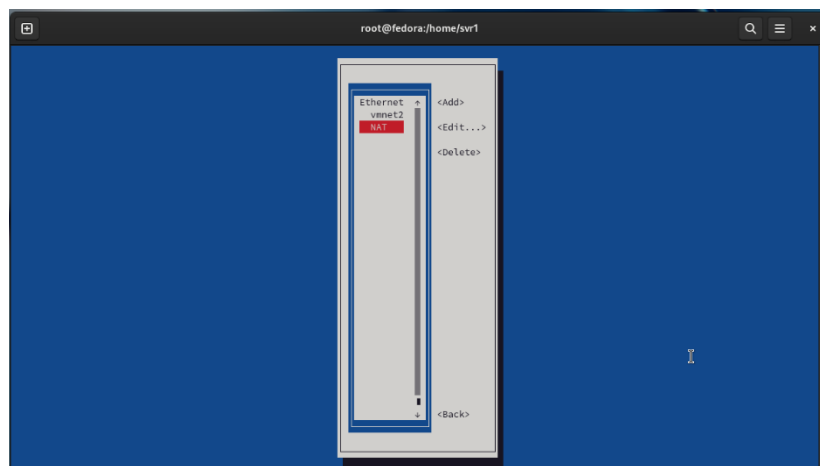


10. You can see two network adapters in the screen, now you can go to the one by one and change like this screenshots(change names also)

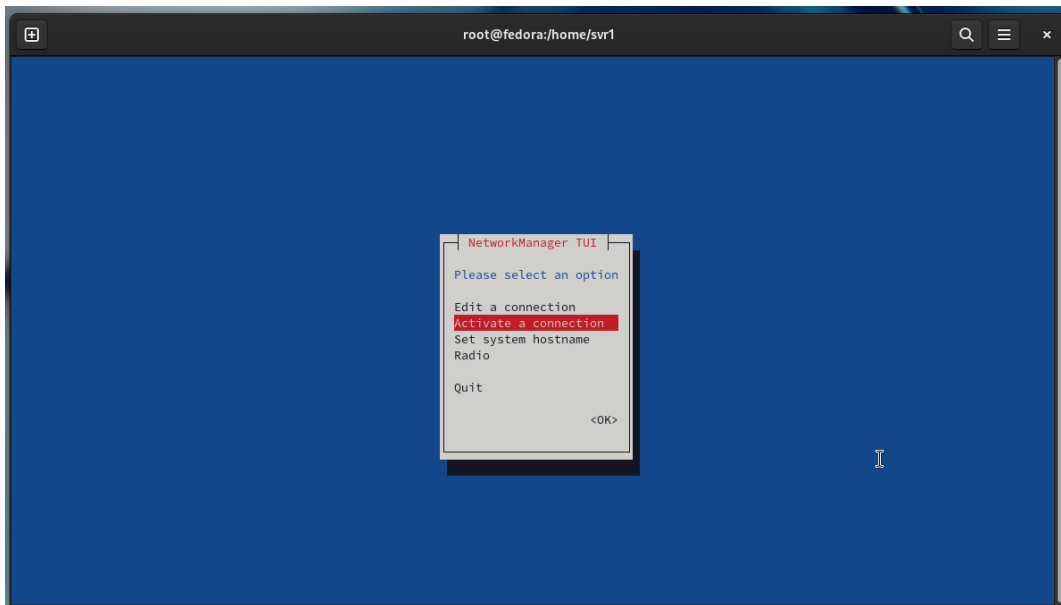




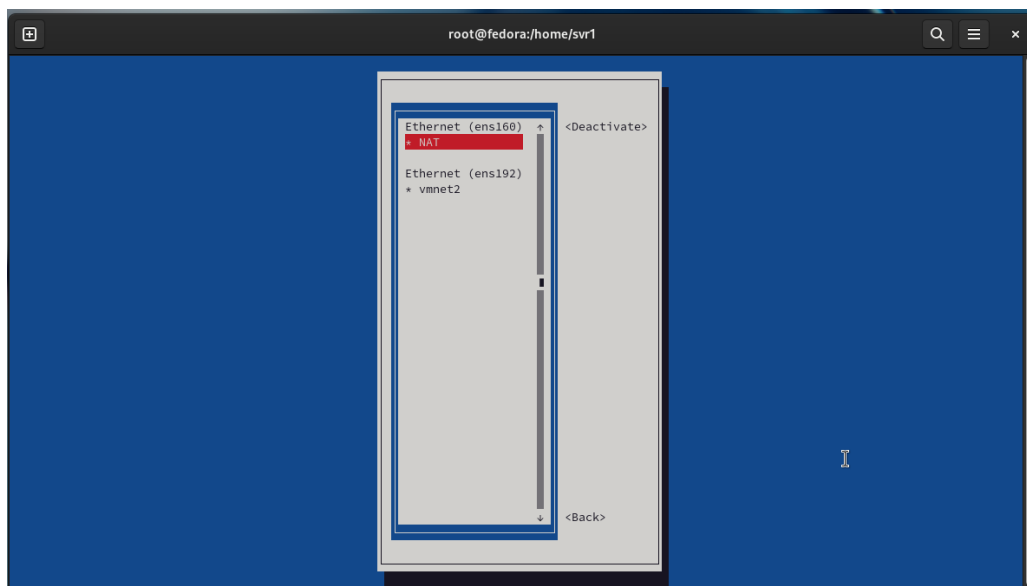
11. You can see changed network adapters



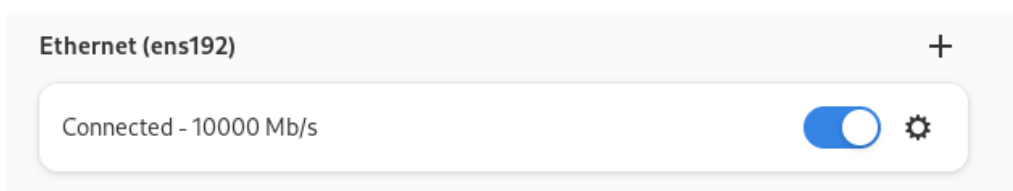
12. After that go to “Activate a connection”



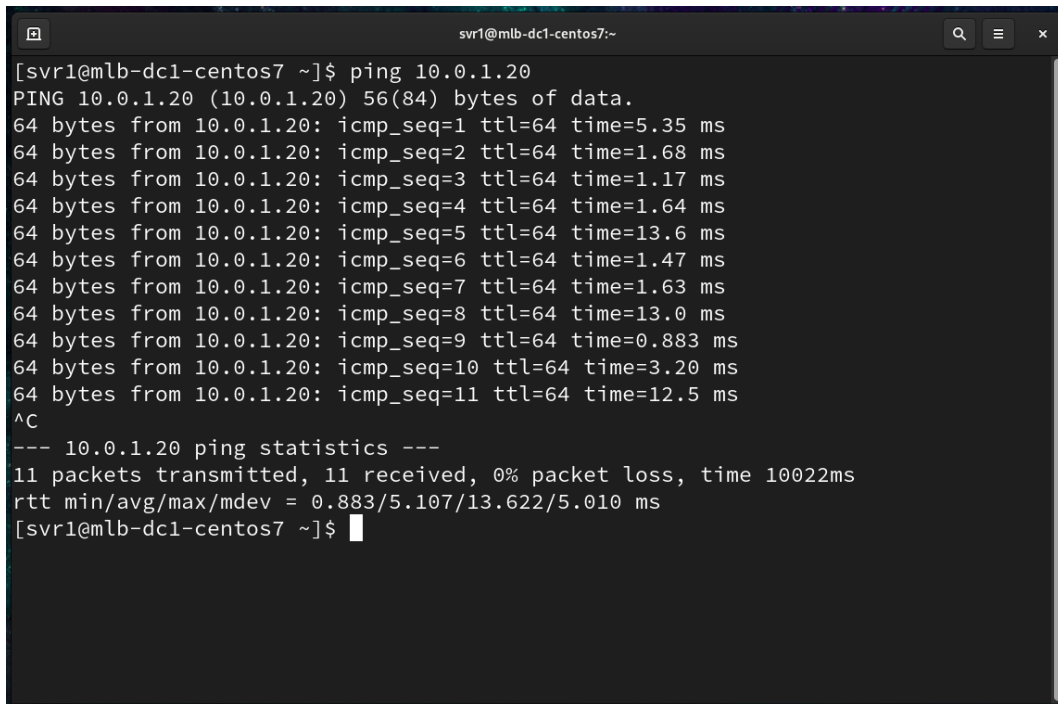
13. Activate two connections, you can see * mark front in the network adapter also activated.



14. Then go to the FedoraOS Settings and It become “Connected Mode”



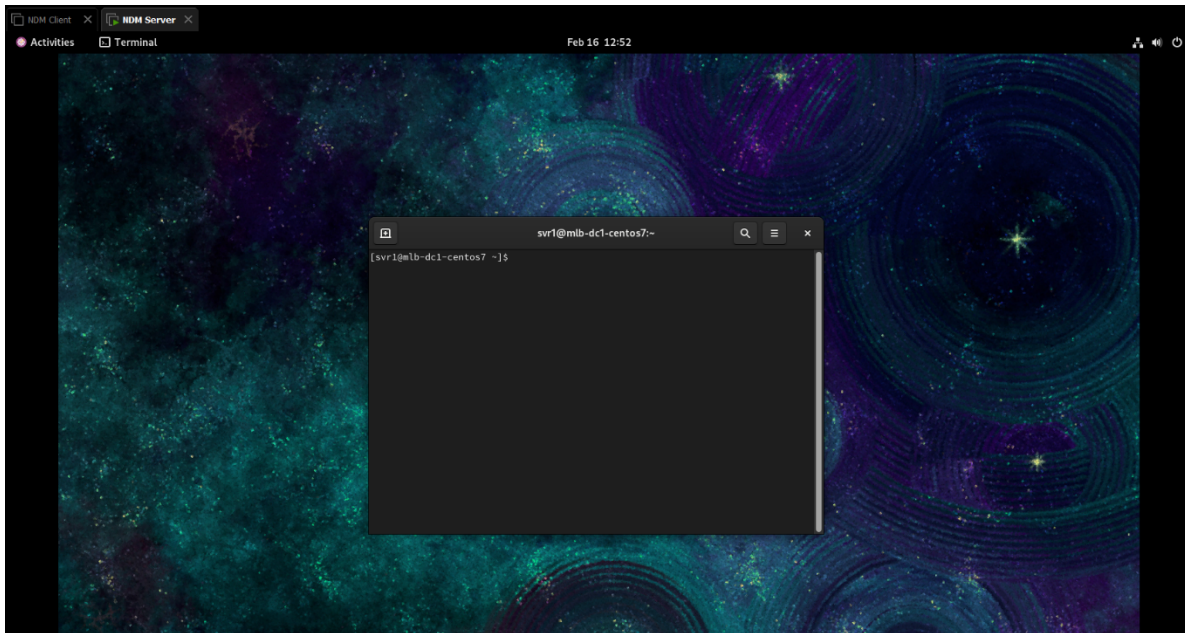
15. Then you can use “ping” command in Server to confirm its work.

A terminal window titled 'svr1@mlb-dc1-centos7:~' showing the execution of the 'ping 10.0.1.20' command. The output displays 11 successful ping responses with varying round-trip times. The command is terminated with Ctrl-C (^C), and the final statistics show 11 packets transmitted and received with 0% packet loss.

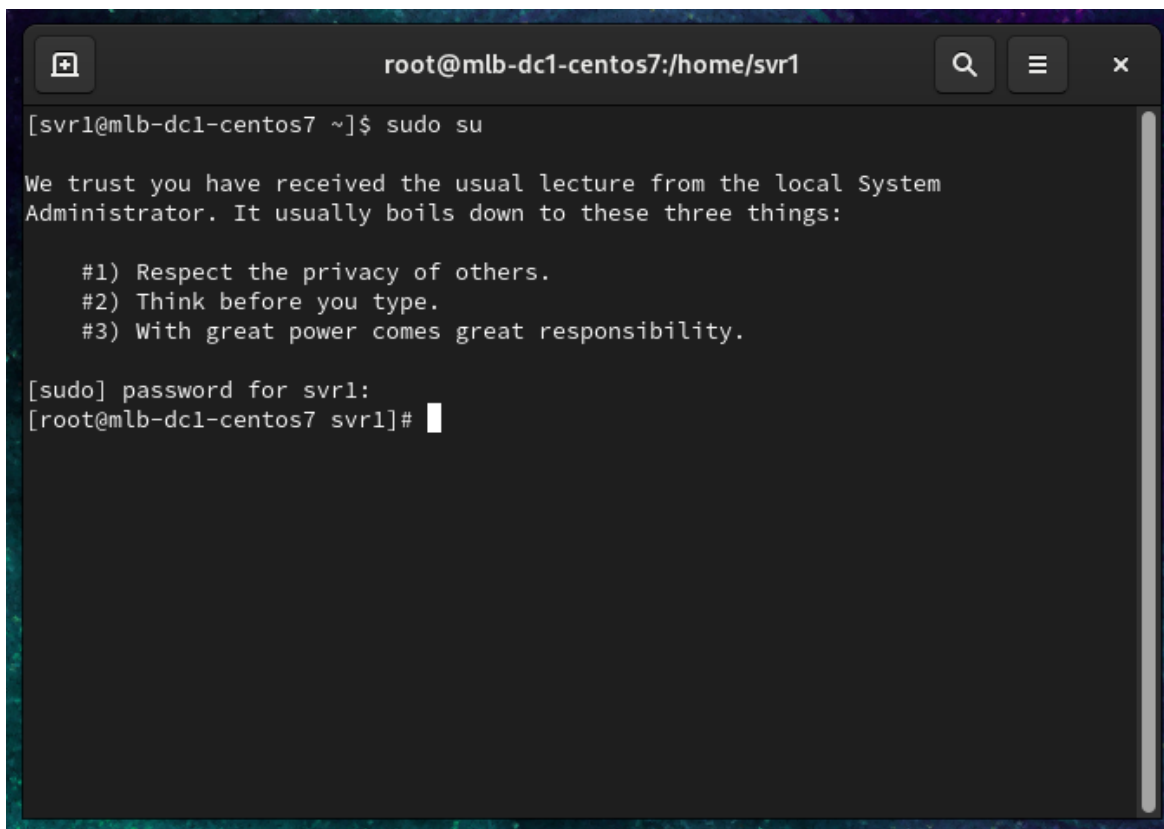
```
[svr1@mlb-dc1-centos7 ~]$ ping 10.0.1.20
PING 10.0.1.20 (10.0.1.20) 56(84) bytes of data.
64 bytes from 10.0.1.20: icmp_seq=1 ttl=64 time=5.35 ms
64 bytes from 10.0.1.20: icmp_seq=2 ttl=64 time=1.68 ms
64 bytes from 10.0.1.20: icmp_seq=3 ttl=64 time=1.17 ms
64 bytes from 10.0.1.20: icmp_seq=4 ttl=64 time=1.64 ms
64 bytes from 10.0.1.20: icmp_seq=5 ttl=64 time=13.6 ms
64 bytes from 10.0.1.20: icmp_seq=6 ttl=64 time=1.47 ms
64 bytes from 10.0.1.20: icmp_seq=7 ttl=64 time=1.63 ms
64 bytes from 10.0.1.20: icmp_seq=8 ttl=64 time=13.0 ms
64 bytes from 10.0.1.20: icmp_seq=9 ttl=64 time=0.883 ms
64 bytes from 10.0.1.20: icmp_seq=10 ttl=64 time=3.20 ms
64 bytes from 10.0.1.20: icmp_seq=11 ttl=64 time=12.5 ms
^C
--- 10.0.1.20 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10022ms
rtt min/avg/max/mdev = 0.883/5.107/13.622/5.010 ms
[svr1@mlb-dc1-centos7 ~]$
```

Server Configuration

1. Go to the CentOS and open terminal



2. Type “sudo su” and go to the root



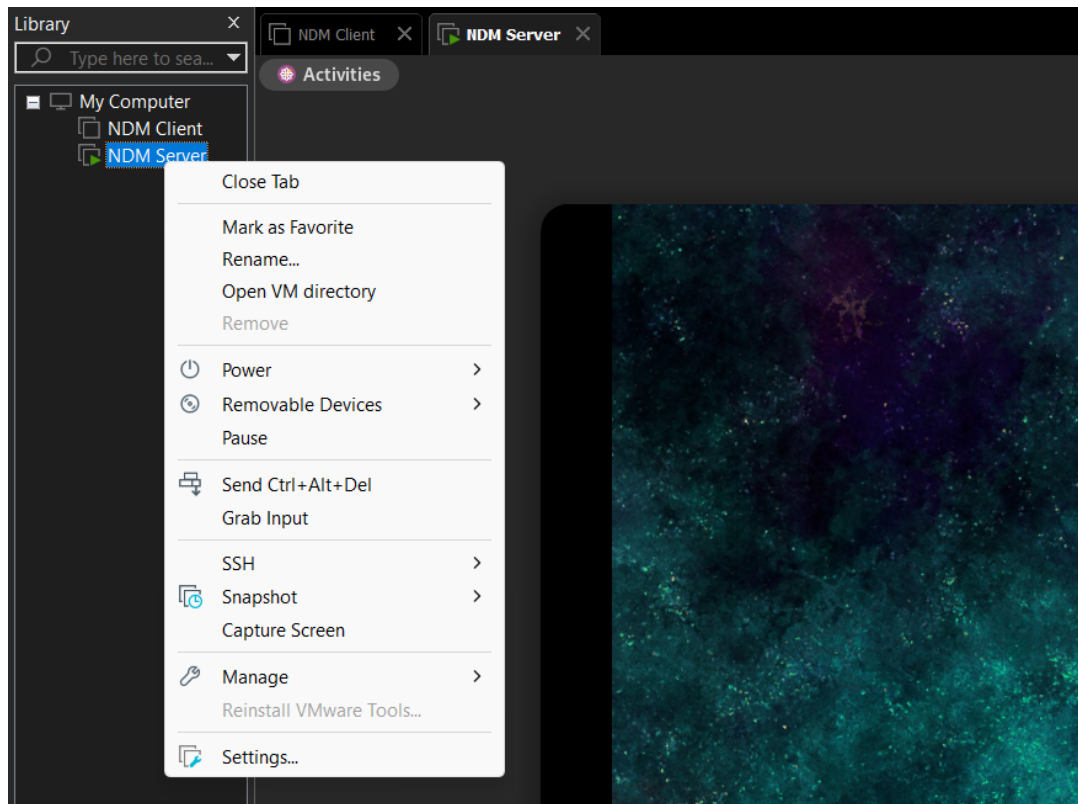
```
root@mlb-dc1-centos7:/home/svr1

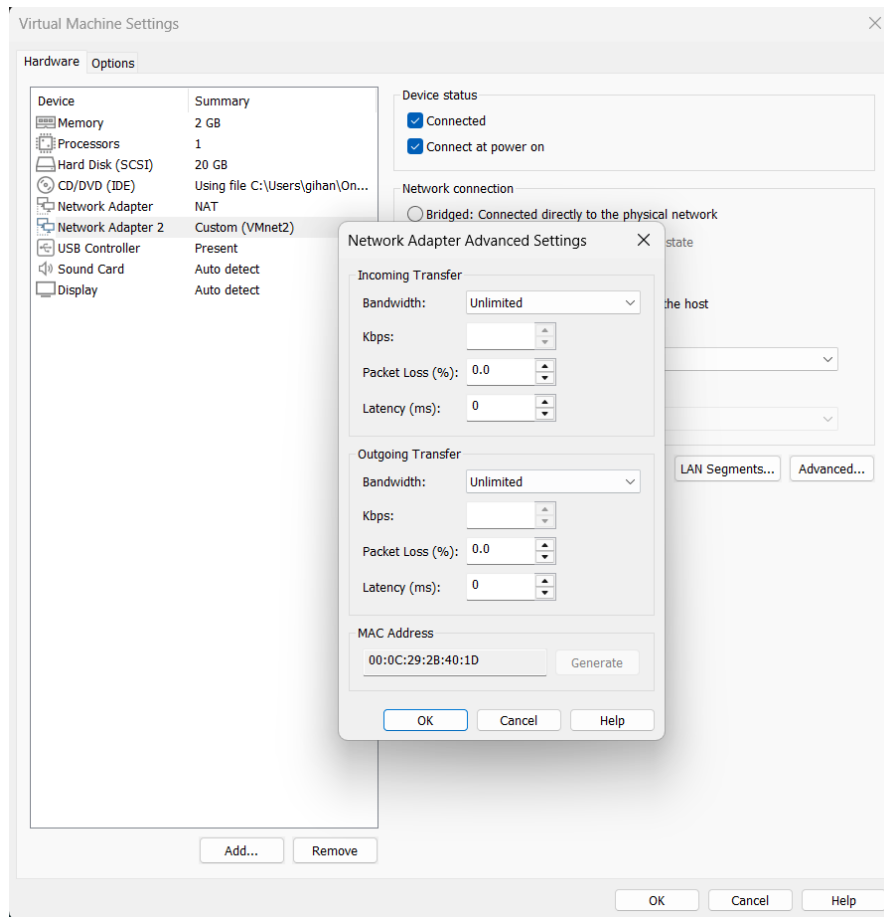
[svr1@mlb-dc1-centos7 ~]$ sudo su
We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

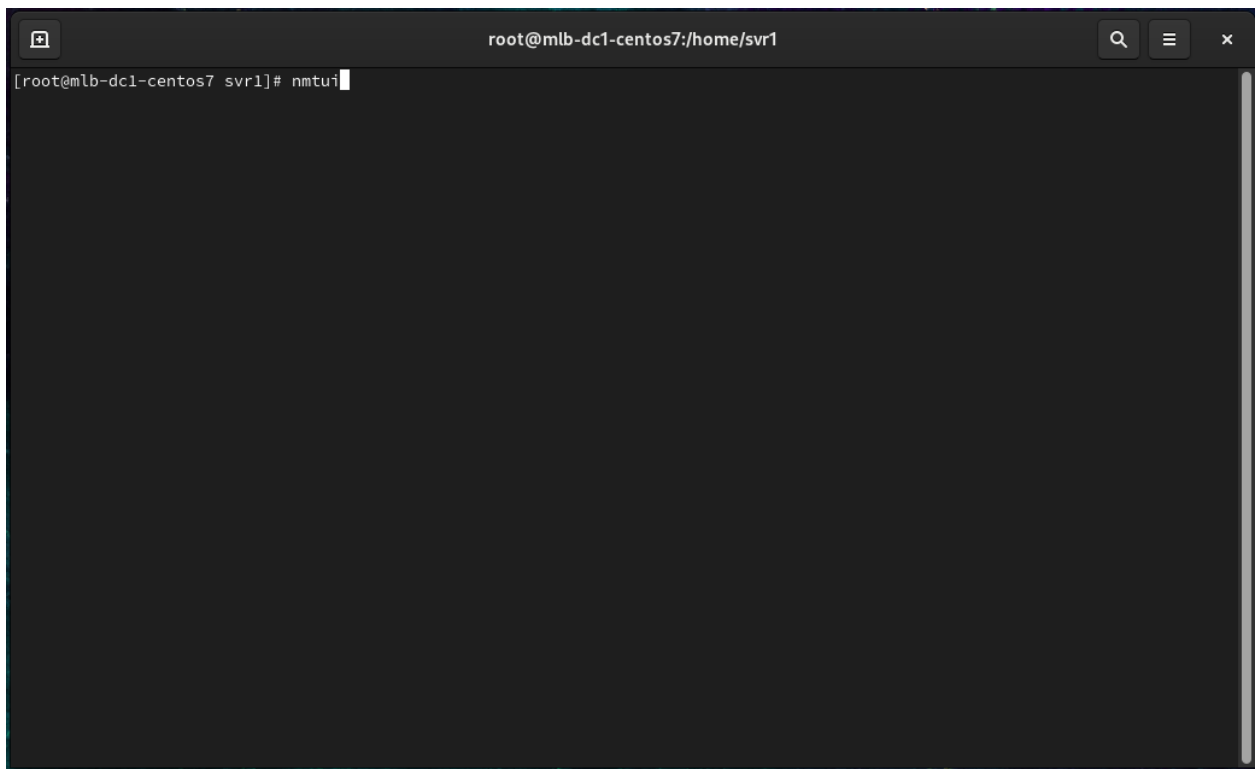
[sudo] password for svr1:
[root@mlb-dc1-centos7 svr1]# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:2b:40:13 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.234.128/24 brd 192.168.234.255 scope global dynamic noprefixroute ens33
        valid_lft 1698sec preferred_lft 1698sec
    inet6 fe80::20c:29ff:fe2b:4013/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens36: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:2b:40:1d brd ff:ff:ff:ff:ff:ff
    altname enp2s4
    inet 10.0.1.2/24 brd 10.0.1.255 scope global noprefixroute ens36
        valid_lft forever preferred_lft forever
    inet6 fe80::1d81:51ca:cc93:340c/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[root@mlb-dc1-centos7 svr1]#
```

4. Go to the Server settings and check the MAC address

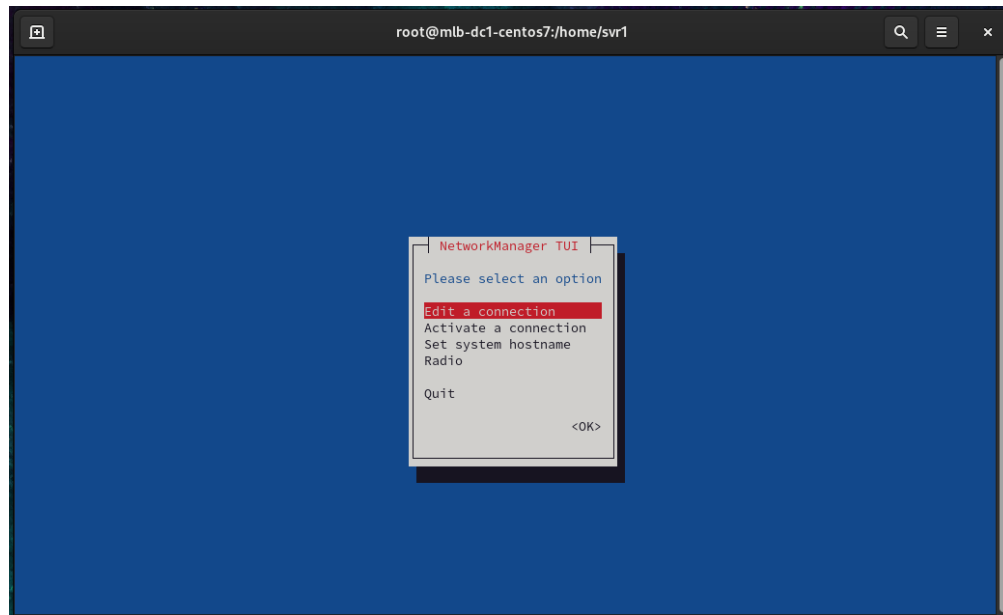




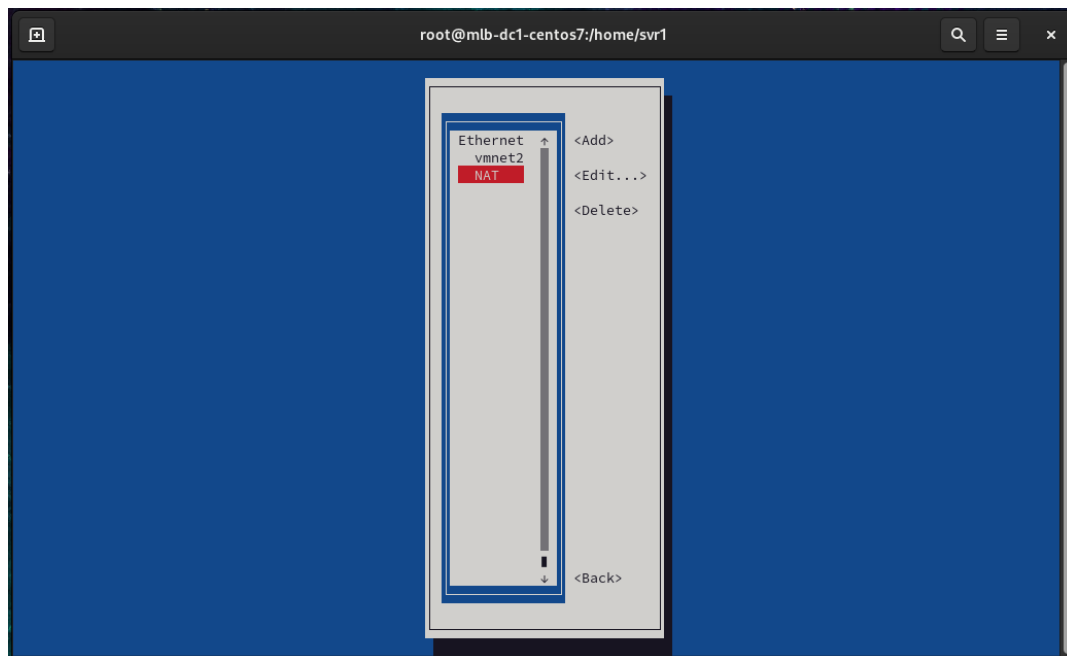
5. Go to the terminal and access the root and enter the “nmtui” Network Manager [follow the 2nd step]



6. Go to the Network Manager TUI and select “Edit a connection”



7. Select the “vmnet2” and change the settings below the screenshot



9. Check using the “ifconfig” and confirm

```
root@mlb-dcl-centos7:/home/svr1

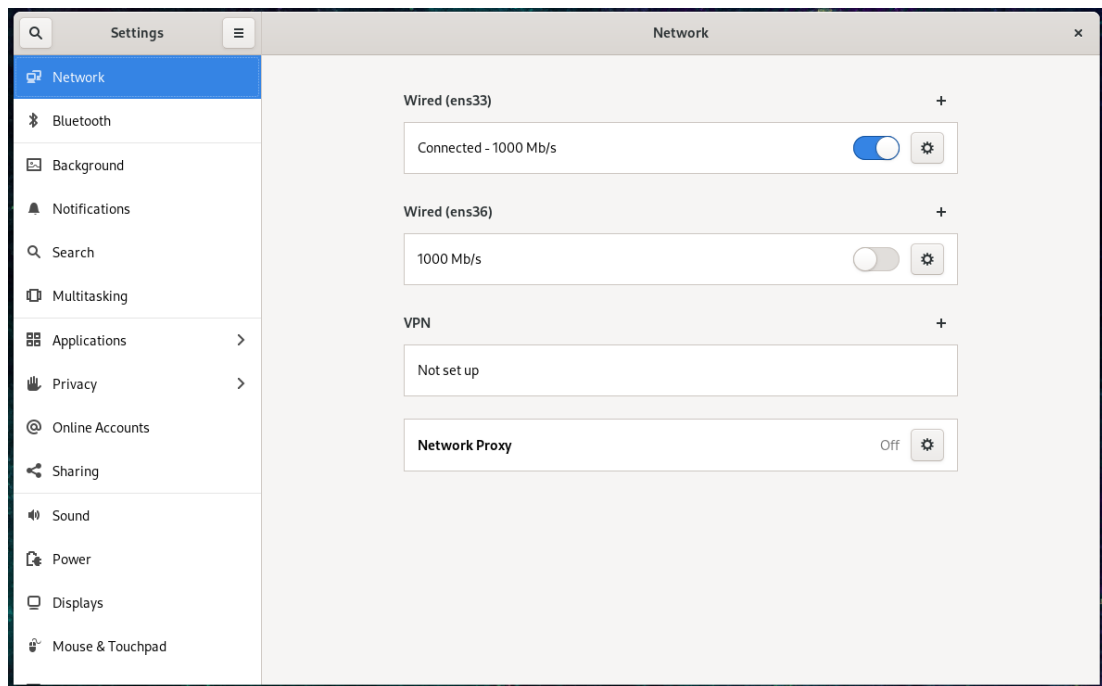
[root@mlb-dcl-centos7 svr1]# ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.234.128 netmask 255.255.255.0 broadcast 192.168.234.255
    inet6 fe80::20c:29ff:fe2b:4013 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:2b:40:13 txqueuelen 1000 (Ethernet)
    RX packets 1702 bytes 926931 (905.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1104 bytes 116889 (114.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ens36: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.1.2 netmask 255.255.255.0 broadcast 10.0.1.255
    inet6 fe80::1d81:51ca:cc03:340c prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:2b:40:1d txqueuelen 1000 (Ethernet)
    RX packets 22 bytes 2226 (2.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 58 bytes 6173 (6.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 24 bytes 2620 (2.5 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 24 bytes 2620 (2.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[root@mlb-dcl-centos7 svr1]#
```

10. Check the CentOS Settings and switch on adapter 2



11. Now check the adapter and it shows “Connected”



12. Ping to the Client and check the connection.

```
svr1@fedora:~  
svr1@ndm-cli-fedora28:~$ ping 10.0.1.2  
PING 10.0.1.2 (10.0.1.2) 56(84) bytes of data:  
64 bytes from 10.0.1.2: icmp_seq=1 ttl=64 time=5.32 ms  
64 bytes from 10.0.1.2: icmp_seq=2 ttl=64 time=13.8 ms  
64 bytes from 10.0.1.2: icmp_seq=3 ttl=64 time=1.41 ms  
64 bytes from 10.0.1.2: icmp_seq=4 ttl=64 time=1.62 ms  
64 bytes from 10.0.1.2: icmp_seq=5 ttl=64 time=1.95 ms  
64 bytes from 10.0.1.2: icmp_seq=6 ttl=64 time=1.34 ms  
64 bytes from 10.0.1.2: icmp_seq=7 ttl=64 time=1.44 ms  
64 bytes from 10.0.1.2: icmp_seq=8 ttl=64 time=4.95 ms  
64 bytes from 10.0.1.2: icmp_seq=9 ttl=64 time=1.42 ms  
^C  
--- 10.0.1.2 ping statistics ---  
9 packets transmitted, 9 received, 0% packet loss, time 8013ms  
rtt min/avg/max/mdev = 1.337/3.691/13.787/3.865 ms  
svr1@ndm-cli-fedora28:~$
```